LIZ HERRERA

Technical Artist — Character Systems & Rigging

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SUMMARY

Character Technical Artist specializing in character rigging systems, OpenUSD pipelines, and Maya/Blender/Houdini tooling. Focused on building production-ready assets and tools that serve downstream Art, Engineering, and ML workflows.

EXPERIENCE

Character Technical Artist | Genies | *Jan 2025 – Present*

Next-Gen Rig Standard

- Designed and shipped the next gen avatar standard, defining the canonical skeleton, OpenXR facial blendshape schema, and layered OpenUSD asset structure used across all production avatars.
- Led the pipeline migration of 1,800+ wearable assets to the new standard using automated refitting and validation tools.

Maya Autorigging Tool + USD Export Pipeline

- Built an end-to-end artist-facing Maya rigging tool covering mesh loading, skeleton binding, deformation setup, validation, and USD export, enabling artists to complete the entire avatar workflow without leaving Maya.
- Designed a modular USD export API producing layered avatar.usd/usdz packages (model, skeleton, skinweights, blendshapes, proxy) with a sparse iteration layer system for non-destructive artist overrides on a published base.

Stack: Python, PySide2/6, OpenUSD, MayaUSD, GitHub Actions API

ML Data Systems — Facial Landmarks + Synthetic Geometry

- Defined the 3D facial landmark specification for humanoid and mammal characters, extending the limited humanoid points to a new dataset used as ML training ground truth.
- Built a Houdini TOPS network generating 6,000+ facial geometry variants from 500 head targets across 5 ethnic groups using zone-based blendshape blending — full batch pipeline.

Stack: *Houdini TOPS, OpenUSD, Python (hou API), OpenCV, NumPy, COCO keypoints*

Supporting Tools

- Material → USD Converter: Maya Python module that batch-converts any shader network to usdPreviewSurface, with on-the-fly Unity MetallicSmoothness → glTF ORM texture conversion using OpenCV/NumPy.
- Blendshape Calibrator: Maya Qt tool for normalizing blendshape target weights with dual list/tree view modes.

Stack: *Python, maya.cmds, MayaUSD, PySide2/6, OpenCV, NumPy*

Character TD | Genies | *Apr 2023 – Dec 2024*

- Provided pipeline tool support and developed rigging tools/pipeline automation
- Implemented rigging and deformation workflows leveraging matrix transforms, vector operations, and skeletal hierarchies for joint orientation compatible with Mixamo Standards.

Stack: Maya, Python

Jr. Character TD | Genies | *Jul 2022 – Apr 2023*

- Built character rigs, wearable assets and props for production

Rigging Artist | Quantic · Raptor Studios | *2020 – 2021*

- Created production character rigs with control systems and deformation setups for animated characters.
- Assisted with character rig setups, skinning, and deformation adjustments; collaborated with animation teams to meet production requirements.

PERSONAL PROJECT

- **MediaPipe → Maya Facial Retargeting:** Used MediaPipe to track ARKit blendshape values from video and map per-frame JSON configs onto a Maya rig's controls — lightweight facial motion retargeting from video to character.

EDUCATION

Engineering in Animation & VFX · Universidad Politécnica de Santa Rosa
Master in Rigging for Animation & Videogames · Libel Academy
Rigging for Character TDs · Keyframe Animation School
Advanced Python for TDs · TD Academy

TECHNICAL SKILLS

DCC: Maya, Houdini (TOPS), Blender, Unreal Engine, Unity, Wrap3D **Languages:** Python, MEL **UI:** PySide2/PySide6, Qt
Pipeline/Systems: OpenUSD, MayaUSD, USDZ, Git, GitHub Actions, Headless DCC pipelines **Rigging:** Skinning, blendshapes, FK/IK, ARKit & OpenXR facial expressions, FACS system, mGear